

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code	CSA0314	Course Title	Data Structures
CO Assessed	CO1	Assessment	Assessment Tool 2: Technical Presentation
Weightage	30% of CIA	Total Marks	100
CO4 Statement	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL3, BL4)</i>		
Student Name	V.Kailash	Reg. No.	192521232
Section/Batch	Slot C /	Date	03-10-2026

Code of Conduct:

I,V.Kailash, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

v.kailash

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Technical Content Accuracy	30%	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood	
Problem Identification	25%	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach	
Use of Tools	20%	Concepts are well-structured hierarchically with logical flow and grouping	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals	
Communication	15%	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides	
Professionalism	10%	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks minimal contribution or ethical awareness	



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An e-commerce website stores product information and frequently updates stock availability.
Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency.

Name : V.Kailash (192521232)

SUBJECT : Data Structures

CODE : CSA0314

GUIDE NAME:

Dr Krishnaraj Mony

Introduction to the Problem

- An e-commerce website manages a large number of products.
- Each product has important information such as:
 - Product ID
 - Product Name
 - Price
 - Category
 - Stock Quantity
- Stock availability changes frequently whenever customers purchase or return products.
- The system must quickly check whether a product is available or out of stock.
- It should also update the stock quantity immediately after every purchase.
- Searching through thousands of products one by one can take more time.

Recommended ADT: Hash Table

- The Hash Table ADT is recommended for managing product information in an e-commerce website.
- A hash table stores data in the form of Key–Value pairs.
- Each product can be identified using a unique Product ID.
- The Product ID acts as the key, while the product details act as the value.
- Example:
 - P101 → Laptop, ₹50,000, Stock: 20
- A hash function converts the Product ID into an index/location in the hash table.
- This allows the system to directly locate the required product instead of checking every product.
- When a customer purchases a product, the stock quantity can be updated quickly.

Operations of Hash Table

A Hash Table supports several important operations required for an e-commerce website. These operations allow product information to be managed quickly and efficiently.

1. Insertion

Used to add a new product to the system.

The Product ID is given as the key.

2. Search

Used to find a product quickly using its Product ID.

The Product ID is passed through the hash function.

3. Update

Used to change product information, especially stock quantity.

Example:

Before purchase $\rightarrow$ Stock = 20

4. Deletion

Used to remove a product that is discontinued or no longer available.

The product is located using its Product ID and removed from the hash table.

How Hash Table Improves Performance Efficiency

Faster Product Search :

- Product ID is used as the key.
- The hash function directly finds the storage location.
- No need to check every product one by one.
- Average search time is $O(1)$.

Quick Stock Updates :

- Stock availability changes frequently after purchases and restocking.
- Hash tables allow the system to directly access the product record.
- Example: P101 → Stock: 20 → Customer buys 3 → Stock: 17.

Efficient Product Addition :

- New products can be inserted quickly using their unique Product ID.
- The system does not need to rearrange the entire product database.

Fast Product Removal :

- Discontinued or unavailable products can be located and deleted efficiently.
- This helps keep inventory information updated.

Conclusion and Key Benefits

The Hash Table ADT is a suitable choice for managing product information in an e-commerce website.

It is especially useful because stock availability is frequently searched and updated.

Product ID can be used as a unique key to identify each product.

Product details such as name, price, category, and stock quantity can be stored as the value.

Key Benefits:

Fast Searching

- Products can be found quickly using their Product ID.
- Average search complexity is $O(1)$.

Fast Stock Updates

- Stock quantity can be increased or decreased quickly after purchases and restocking.

Efficient Data Management

- Products can be easily inserted, updated, searched, and deleted.

THANK YOU